Supplementary Information for:

GAB-Seg: A Task-Specific Boundary-Aware Framework for Irregular Multi-scale Spot-like Target Segmentation in Anterior Segment Images

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Supplementary Methods: Baseline Training Protocols and Hyperparameter Configurations

To ensure fairness and reproducibility, all baseline models evaluated in Table 1 were trained from scratch (randomly initialized without any external pre-training) using identical data splits. Unless otherwise specified, the models were trained for 200 epochs (400 epochs for GenSeg) utilizing a hybrid BCE-Dice loss function. Automatic Mixed Precision (AMP) was enabled across all experiments (except for the standard U-Net) to accelerate training. The specific hyperparameters and learning rate schedules were configured following the default or optimal settings of their respective architectures.

- U-Net: We utilized the standard Adam optimizer with a constant learning rate of 1×10^{-4} and a batch size of 16, without a learning rate scheduler.
- Attention-UNet, EGE-UNet, & VM-UNet: We employed the AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1 \times 10^{-8}$) with a base learning rate of 1×10^{-3} and a batch size of 8. The weight decay was set to 0.1 for Attention-UNet, and 0.01 for both VM-UNet and EGE-UNet. A Cosine Annealing scheduler was applied with $T_{max} = 50$ and $\eta_{min} = 1 \times 10^{-5}$. In addition, official multi-scale deep supervision (with branch weights of 0.1, 0.2, 0.3, 0.4, and 0.5) was enabled for EGE-UNet.
- Swin-UNet: Strictly following the official configuration, we utilized the AdamW optimizer with a batch size of 8, a base learning rate of 5×10^{-4}, and a weight decay of 0.05. A Cosine learning rate scheduler with a 20-epoch linear warmup (warmup initial LR = 5×10^{-7}) and a minimum learning rate of 5×10^{-6} was applied over the 200-epoch training cycle.
- GenSeg: As our foundational baseline, its training protocol and hyperparameter settings were kept completely identical to those of the proposed GAB-Seg in the main manuscript (trained for 400 epochs with a batch size of 8 using the AdamW optimizer, a base learning

rate of 5×10^{-4}, and a Cosine Annealing scheduler).

Supplementary Note: Complete Multi-Metric Target Size-Stratified Evaluation

To provide a comprehensive multi-scale evaluation beyond the overall and size-stratified Dice scores reported in Section 3.5 (Table 5) of the main manuscript, this section presents the complete quantitative breakdown across all five evaluation metrics—Dice, Intersection over Union (IoU), 95% Hausdorff Distance (HD95), Precision, and Recall—on the test set. Based on the intrinsic pixel-area distribution of the lesion spots at the standardized 256×256 input resolution (median area = 306 pixels; mean area = 526.7 pixels), the multi-metric results for minute targets (300–600 pixels, occupying 0.45%–0.91% of the total image area) and relatively large targets (>600 pixels, occupying >0.91% of the total image area) are detailed in Supplementary Table S1 and Supplementary Table S2, respectively. Both the comparative baseline models and the proposed GAB-Seg framework trained under varying sample sizes (20, 40, and 60 training images) are included for a thorough comparison.

Supplementary Table S1. Complete multi-metric performance evaluation on minute targets (300–600 pixels) on the test set.

Model	Dice↑	IoU↑	HD95↓	Precision↑	Recall↑
Attention-UNet	0.3962	0.2947	144.57	0.3479	0.5188
Swin-UNet	0.4312	0.3319	110.91	0.5448	0.4700
EGE-UNet	0.1896	0.1314	229.62	0.2805	0.1661
VM-UNet	0.4304	0.3329	116.35	0.4404	0.5235
GenSeg	0.4032	0.3131	126.53	0.4165	0.5415
GAB-Seg (20 Images)	0.4247	0.3452	140.04	0.4775	0.3979
GAB-Seg (40 Images)	0.6401	0.5101	45.67	0.6916	0.6507
GAB-Seg (60 Images)	0.5661	0.4620	91.54	0.5260	0.6695

Supplementary Table S2. Complete multi-metric performance evaluation on relatively large targets (>600 pixels) on the test set.

Model	Dice↑	IoU↑	HD95↓	Precision↑	Recall↑
Attention-UNet	0.6116	0.4808	47.23	0.7046	0.6240
Swin-UNet	0.5654	0.4384	45.59	0.7063	0.5678
EGE-UNet	0.2199	0.1488	164.34	0.5166	0.1798
VM-UNet	0.6502	0.5106	34.72	0.7302	0.6730
GenSeg	0.5651	0.4350	61.76	0.6321	0.6121
GAB-Seg (20 Images)	0.5268	0.4001	31.78	0.8806	0.4470
GAB-Seg (40 Images)	0.6616	0.5315	31.81	0.8377	0.5982
GAB-Seg (60 Images)	0.6955	0.5616	20.85	0.7592	0.7122

Note on Ultra-Minute Targets (<300 pixels): As analyzed in Section 3.5 and demonstrated in Table 5 of the main manuscript, detailed multi-metric breakdowns for the ultra-minute target tier (<300 pixels, accounting for <0.45% of the image area) are omitted here because both GAB-Seg and all comparative baseline models encounter a shared physical resolution bottleneck and fail to achieve valid segmentation in this regime (with Dice scores approaching zero across all models, ranging merely from 0.0000 to 0.0591). Specifically, given the 256×256 input resolution, five consecutive downsampling stages compress ultra-minute spots (equivalent side length <17 pixels) to less than a single pixel in the deepest feature maps, leading to severe attenuation of edge gradients and spatial topology. Consequently, secondary metrics (such as Precision, Recall, and penalized HD95 values) in this tier merely reflect sporadic responses to background noise rather than meaningful segmentation performance, and are therefore not separately tabulated.